CPHEEO Water Supply Design Criteria & Technical Notes

Comprehensive Summary of Standards, Empirical Formulas, Hydraulic Parameters & O&M; Norms (2024 Guidelines)

> **Executive Summary & Regulatory Framework:** This document synthesizes key engineering criteria, hydraulic equations, and unit design parameters from the **CPHEEO Manual on Water Supply and Treatment Systems (2024 Edition)** and Jal Jeevan Mission (JJM) / Drink-from-Tap (DfT) Mission directives. It covers planning horizons, demand forecasting, pipeline hydraulics, pumping machinery selection, WTP unit processes, valve siting/spacing, and pipe laying standards.

1. Planning Horizons, Per Capita Demand & Loss Benchmarks

Infrastructure Component	Design Period	Per Capita / Design Norm	Operational Criteria
Raw & Treated Water Intakes	30 Years	Full Ultimate Demand	Designed for ultimate stage flow
Water Treatment Plant (WTP)	30 Years	20% Hydraulic Overload	Interconnecting channels handle +20%
Raw & Treated Pumping Mains	30 Years	22 Hours Pumping/Day	Velocity 0.8 to 1.2 m/s
Pumping Machinery & Motors	15 Years	Intermediate Stage Demand	Re-evaluated at 15-year horizon
Elevated Service Reservoirs (ESR)	30 Years	Min 33% Daily Demand	Mass balance or 33% ultimate demand
Urban Distribution Network	30 Years	135 - 150 LPCD	Peak Factor 2.5 (continuous 24x7 DfT)
Rural Distribution (JJM)	30 Years	55 LPCD	Peak Factor 3.0 (terminal head 7m / 12m)
Floating Population	Current	45 LPCD (Bathing)	25 LPCD without bathing facilities

Key Demand & NRW Formulas:

• **Firefighting Demand Formula:** $Q = 100\sqrt{P}$ in m³/day (where P = population in thousands). Fire demand is added proportionately across operational zones.
• **Non-Revenue Water (NRW) Targets:** WTP process loss ≤ 3%; Distribution system losses ≤ 10% (under 24x7 100% metered DfT schemes); Groundwater direct pumping schemes total loss ≤ 11%; Overall national utility NRW target ≤ 15%.

2. Pipeline Hydraulics, Friction Equations & Water Hammer

Hydraulic design of pressure conduits and gravity mains relies on standard friction head loss formulas. The Hazen-Williams formula is universally used for pressurized water supply design in India.

Formula / Parameter	Mathematical Expression	Recommended CPHEEO Coefficients / Value
Hazen-Williams Velocity	V = 0.849 · C · R^0.63 · S^0.54	C = 140 (DI K-9, Mortar Lined Steel, HDPE, uPVC) C = 100 (Unlined CI/MS pipes)
Hazen-Williams Head Loss	h_f = (10.68 · L · Q^1.852) / (C^1.852 · D^4.87)	L, D in meters; Q in m³/s; h_f in meters
Darcy-Weisbach Equation	h_f = (f · L · V²) / (2 · g · D)	Roughness k = 0.03 mm (Plastic), 0.035 mm (Lined metallic)
Water Hammer Celerity	c = 1425 / √(1 + (K_w/E_p)·(D/t))	c = 800 - 1200 m/s for DI/Steel mains

Formula / Parameter	Mathematical Expression	Recommended CPHEEO Coefficients / Value
Surge Pressure Rise	h_surge = (c · V_0) / g	Critical closure time t_c = 2L / c
Transmission Velocity	Recommended Range: 0.8 - 1.2 m/s	Optimum design velocity = 1.0 m/s; Max = 2.5 - 3.0 m/s
Distribution Velocity	Recommended Range: 0.6 - 1.5 m/s	Minimum velocity = 0.3 m/s (to avoid siltation)
Residual Nodal Pressure	Class I & II Cities: 17 - 21 m	Other towns: 12 - 15 m; Rural JJM: 7 m (no FCV), 12 m (with FCV)

Surge Protection & Water Hammer Control Devices:

1. **Air Vessels:** Installed near pump discharge to cushion downsurge and suppress upsurge.
2. **Surge Anticipation Valves (SAV):** Sensor-actuated valves that open on power failure / initial pressure drop to vent return upsurge.
3. **Air Cushion Valves (ACV):** Admits large volumes of air during downsurge and expels air under controlled resistance during column return.
4. **Zero Velocity Valves (ZVV):** In-line spring-loaded discs that close forward water columns at zero momentum before reverse velocity begins.

3. Pumping Station Design & Pump Machinery Selection

Parameter / Component	CPHEEO Standard Criteria	Engineering Notes & Operational Limits
Operating Hours	20 to 22 Hours/Day	Allows 2 to 4 hours for maintenance and power outages
Specific Speed (N_s)	38 to 68 (SI) / 2,000 to 3,500 (USCU)	Optimum efficiency range; N_s must never exceed 80 (SI)
Operating Speed (RPM)	980 RPM Preferred for Large Pumps	1480 RPM acceptable; 2900 RPM strictly restricted to borewells
Column Pipe Velocity	1.75 to 2.75 m/s	Lower velocity for small Q, higher for large Q
Bell Mouth Velocity	1.20 to 1.40 m/s Entrance Velocity	Bell mouth diameter = 1.5 to 1.8 times suction pipe diameter
Pump Power Formula	bkW = (g · Q · H · Spg) / eff_p	g = 9.81 m/s²; Q in m³/s; H in m; eff_p = pump efficiency
Motor Rating Margin	15% to 20% Reserve Margin	Prevents motor overloading during HWL / LWL operating shifts
Standby Pump Capacity	50% to 100% Standby Provision	1W+1S (100%), 2W+1S (50%), or 3W+1S (33%)

Variable Frequency Drives (VFD) & Affinity Laws: VFD pumps regulate rotational speed (N) to match varying network demand.
• Discharge: $Q_2 / Q_1 = N_2 / N_1$ | • Head: $H_2 / H_1 = (N_2 / N_1)^2$ | • Power: $P_2 / P_1 = (N_2 / N_1)^3$

4. Water Treatment Plant (WTP) Unit Operation Design Criteria

WTP process units are sized for rated design flow (30-year horizon). Interconnecting conduits, channels, and weir drops are designed for **20% hydraulic overloading**. Total hydraulic head loss through WTP (cascade aerator lip to CWR FSL) ≤ **5.0 m**.

Unit Operation	Key Design Parameter	CPHEEO Recommended Range / Value	Dimensional & Velocity Limits
Cascade Aerator	Area Loading Rate Drop & Steps	0.015 - 0.045 m²/(m³/hr) 4 to 6 steps; 0.60 - 1.50 m drop	Step rise 0.15-0.25 m; tread ≥ 2x rise Central shaft velocity < 0.60 m/s
Mechanical Flash Mixer	Detention Time (t) Velocity Gradient (G)	t = 60 seconds G = 300 s^{-1} (m/sec/m)	Paddle tip velocity = 3 - 4 m/s Tank depth-to-diameter ratio = 1:1 to 3:1
Hydraulic Weir Mixer	Freefall Drop Velocity Gradient (G)	Drop = 0.40 - 0.60 m G = 800 - 1000 s^{-1}	Detention time in d/s channel = 2 - 5 sec Channel velocity = 1.0 - 1.5 m/s
Mechanical Flocculator	Detention Time (t) Gradient & G·t Product	t = 20 - 40 minutes G = 40 - 70 s^{-1}; G·t = $2x10^4$ to $6x10^4$	Paddle tip velocity = 0.40 - 0.70 m/s Paddle area = 10% - 25% of cross-section

Unit Operation	Key Design Parameter	CPHEEO Recommended Range / Value	Dimensional & Velocity Limits
Clariflocculator	Surface Loading Rate (SLR) Side Water Depth (SWD)	SLR = 30 - 40 $m^3/(m^2 \cdot day)$ SWD ≥ 3.50 m; Detention 2 - 3 hrs	Weir loading = 200 - 300 $m^3/(m \cdot day)$ Flocculated water inlet velocity ≤ 0.005 m/s
Tube Settler Module	Surface Loading Rate (SLR) Module Geometry	SLR = 5 - 6 $m^3/(m^2 \cdot hr)$ [5000-6000 LPH/m^2] 50x50 mm tubes, 600 mm long, 60° inclination	SWD = 3.0 - 4.0 m above hoppers Detention time = 30 - 45 min; Weir load ≤ 400 $m^3/m/d$
Rapid Sand Gravity Filter	Filtration Rate Media Depth & Size	Rate = 5 - 6 $m^3/(m^2 \cdot hr)$ [5000-6000 LPH/m^2] Sand depth = 0.60 - 0.80 m (d_{10} = 0.55-0.60 mm)	Uniformity Coefficient (UC) = 1.5 - 1.7 Graded supporting gravel depth = 0.45 - 0.60 m
Filter Backwash System	Backwash Water Rate Air Scour Rate	Water Wash = 35 - 45 $m^3/(m^2 \cdot hr)$ [8-10 min] Air Scour = 35 - 45 $m^3/(m^2 \cdot hr)$ at 0.35-0.40 kg/cm^2	Backwash tank head = 8.5 m + pipe losses Air piping velocity ≤ 25 m/s
Chlorine Contact Tank	Detention Time Free Residual Chlorine	Detention time ≥ 30 minutes 4 to 6 baffling channels	Free residual chlorine at consumer tap ≥ 0.2 mg/L Super-chlorination during epidemics ≥ 2.0 mg/L

5. Valves Siting, Sizing & Spacing Standards

Valve Type	Functional Objective	Sizing Criteria (d vs D)	Siting & Spacing Standards
Air Release Valve (Small Orifice)	Release entrained air pockets during normal flow	Inlet 12 - 50 mm; Orifice 1 - 10 mm	At summits, grade reductions, and every 500-1000m on long slopes
Kinetic Air Valve (Double Chamber)	Large air venting during fill; vacuum protection	1/4 to 1/5 of main pipe diameter D (Ratio 1:8 vacuum, 1:12 release)	At major summits, hydraulic grade peaks, and d/s of sectionalizing valves
Scour / Washout Valve	Drain pipe section and flush accumulated silt	d = 0.4D to 0.6D (80mm for D≤300; 100mm for D≤600; 150mm for D≤1000; 200mm for D>1000)	At all pronounced valleys/dips (dip > 10m relative to adjacent crests)
Line Isolation Valve (Sluice/Butterfly)	Sectionalize pipeline for maintenance/repairs	Full bore or slightly reduced with taper	Spaced 1 to 5 km apart (1 km in urban networks, 3 km in cross-country)
Pressure Reducing Valve (PRV)	Maintain constant preset d/s pressure	Sized based on flow rate & cavitation factor	In steep/hilly terrain to limit static heads to < 60m
Head-Dissipating Manifolds	Dissipate high hydraulic head on steep drops	Parallel pipe-valve assembly or orifice series	Spacing between adjacent head-dissipating units ≥ 100 x D

6. Pipe Laying, Trench Cover & Structural Anchorages

Parameter	In Ground (Buried in Trench)	On Ground (Exposed on Supports)	Hilly Terrain Special Criteria
Trench Geometry	Width = OD + 400 to 600 mm (200 mm clearance each side)	Supported on concrete pedestals / saddles	Rock excavation 100-150 mm below invert; replace with sand bed
Minimum Cover	0.60 m in non-traffic / footpaths 1.00 m under heavy traffic roads	Exposed; clamped with M.S. straps on piers	0.90 - 1.20 m cover to prevent freezing (below local frost line)
Maximum Cover	2.50 to 3.00 m Maximum Depth	N/A	Cover > 3.0 m requires concrete cradling and structural load calculation
Pipe Orientation	Socket/bell ends face UPSTREAM / gradient	Socket ends face UPSTREAM	Prevents joint slippage during backfilling on steep slopes
Slope Anchorage	Concrete anchor blocks at bends/dead ends	Pedestals with integral concrete keys in rock	Mandatory anchor blocks for slopes steeper than 1 in 6 (16.7%)

Thrust Block Design Criteria (IS: 5330):
• Hydrostatic Thrust Force at Bends: $F = 2 \cdot P \cdot A \cdot \sin(\theta / 2)$ (where P = design pressure, A = pipe area, θ = deflection angle).
• Thrust at Reducers: $F = P \cdot (A_1 - A_2)$ | • Safety Factor against sliding/overturning ≥ **1.5**.
• Minimum Reinforcement: **5 kg/m²** at maximum spacing of **500 mm c/c**.